Dimensionality Reduction on AwA2 Deep Features: Linear SVM with Feature Selection, PCA, and Sparse Coding

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Abstract

*This report investigates dimensionality reduction for image classification on the Animals with Attributes 2 (AwA2) benchmark. We start from 2048-dimensional ResNet-101 features and evaluate a linear Support Vector Machine (SVM) baseline, then apply three complementary dimensionality-reduction strategies—ANOVA-based feature selection (*SELECTKBEST*), Principal Component Analysis (PCA), and sparse coding with OMP—across a wide range of target dimensions. The baseline linear SVM achieves **92.52%** test accuracy on the full 2048-d representation. After dimensionality reduction, none of the three methods surpasses the baseline; the best reduced representation (SelectKBest at 768-d) reaches **92.16%**, showing only a small performance drop with substantial compression. Systematic experiments reveal that PCA provides the best trade-off between accuracy and compression, while sparse coding is consistently the weakest method under our fast OMP configuration. We identify the optimal method and dimensionality for this dataset and provide detailed analyses of performance variance across the full dimensionality spectrum.*

1. Experimental Setting

Dataset. We use the Animals with Attributes 2 (AwA2) benchmark [1, 2], a widely used dataset for zero-shot and fine-grained recognition. AwA2 contains **37,322 images** spanning **50 animal categories** with rich semantic variation (from common mammals to marine species). For each image, the dataset provides pre-extracted **2048-dimensional ResNet-101** [3] feature vectors obtained from the penultimate layer of a network pre-trained on ImageNet. These deep features serve as the sole input representation throughout all experiments; raw pixels are never used.

Train/test split. Following the project specification, images within each of the 50 classes are split independently into **60% training** and **40% testing**, using a fixed random seed (42) for reproducibility. This per-class stratified split yields **22,373 training samples** and **14,949 test samples**. Class-level split sizes range from 40 (minimum) to 787 (maximum) test samples, reflecting the natural class-size imbalance in AwA2. All dimensionality-reduction methods and classifier hyperparameter searches operate exclusively on the training split; the test set is never used during any fitting step.

Preprocessing. Before applying any dimensionality-reduction method we fit a `StandardScaler` on the training features and apply the same transformation to the test features. Zero-centering and unit-variance normalisation ensures that features with large magnitudes do not dominate the ANOVA F-statistics, PCA covariance computation, or sparse-coding objective.

Classifier. Across all experiments (baseline and dimensionality-reduced variants) we use a **linear SVM** implemented by `LinearSVC` [4, 5] from scikit-learn [6]. For Q3/Q4, to keep runtime manageable across many method-dimension settings, we reuse the best value from Q1/Q2 ($C = 0.01$) for all LinearSVC fits instead of re-running cross-validation per setting. The solver uses the primal formulation (`dual=False`) with a maximum of 5000 iterations.

2. Baseline: Linear SVM on Full Features

2.1. Setup

For the baseline (Question 2), the full 2048-d ResNet-101 features are fed directly into a linear SVM without any dimensionality reduction. Cross-validation (5-fold, same grid as above) selects $C^* = 0.01$.

2.2. Results

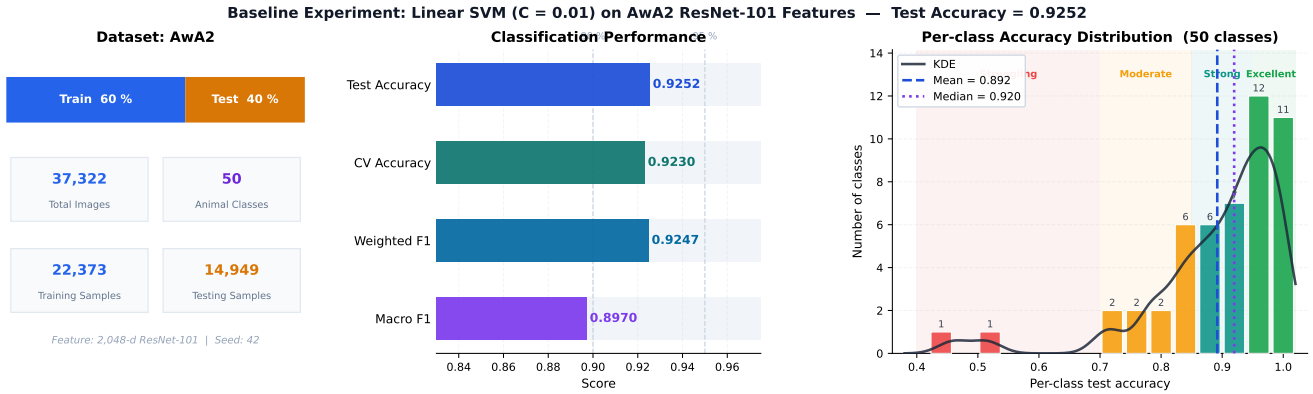

Figure 1. Baseline experimental overview. *Left*: class-wise 60/40 train/test split sizes. *Middle*: CV accuracy, test accuracy, macro F1, and weighted F1. *Right*: distribution of per-class test accuracy across 50 classes.

The baseline achieves a test accuracy of **92.52%**, with macro F1 of 89.70% and weighted F1 of 92.47% (Figure 1). The gap between macro and weighted F1 indicates that a small number of difficult classes pull the unweighted average down.

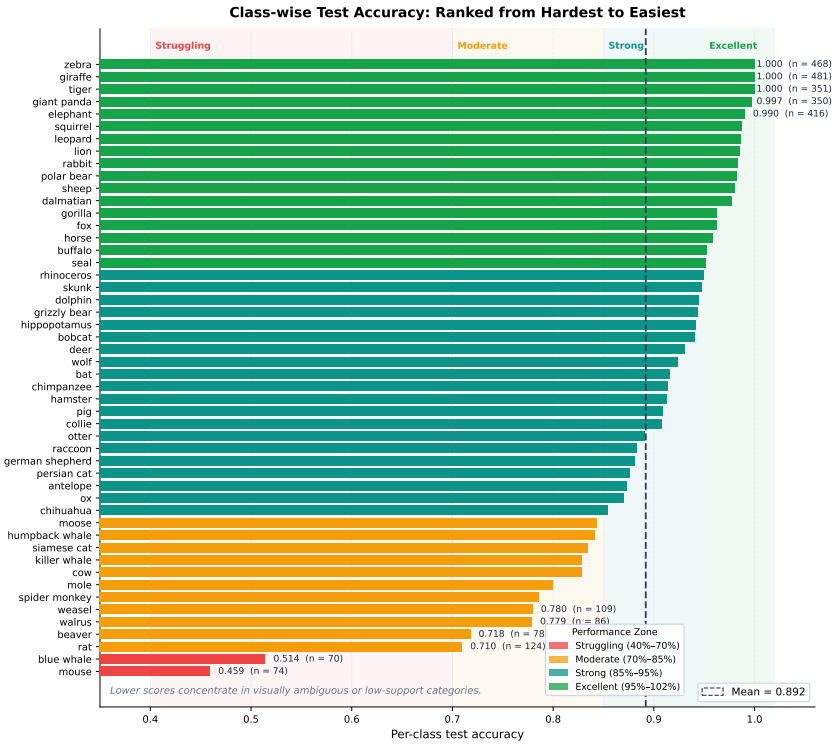

Figure 2. Per-class test accuracy ranked from hardest to easiest. 30 of 50 classes exceed 90% accuracy; the bottom cluster is dominated by visually ambiguous small mammals.

Figure 2 shows that *tiger*, *giraffe*, and *zebra* reach **100%** accuracy, while *mouse* (45.95%) and *blue whale* (51.43%) are the most difficult. The support-vs-accuracy scatter (Figure 3) reveals a positive Pearson correlation of **0.624**, confirming that classes with fewer test samples are more susceptible to misclassification.

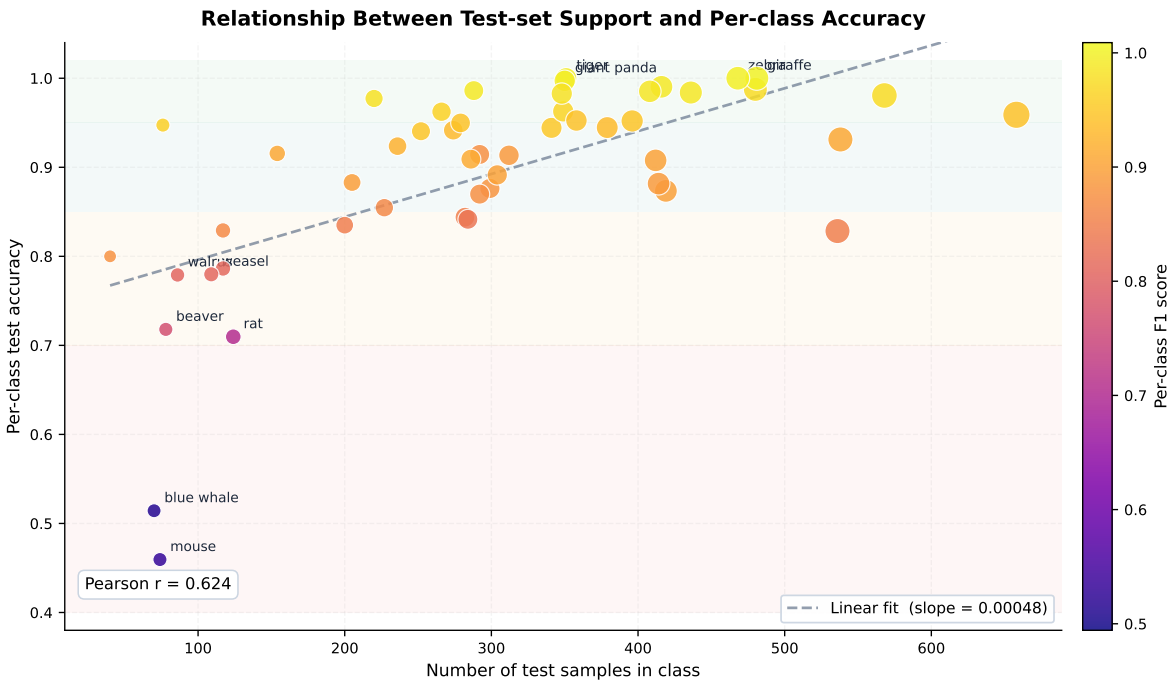

Figure 3. Relationship between test-set support and class accuracy. Colour encodes per-class F1. Classes with more samples tend to achieve higher, more stable accuracy.

These results establish that ResNet-101 features are already highly discriminative for AwA2, and a simple linear decision boundary exploits them effectively. The remaining error is concentrated in a handful of visually confusable or low-support categories.

3. Dimensionality Reduction Methods

We evaluate three fundamentally different strategies for reducing the 2048-d feature vectors to a lower-dimensional representation.

3.1. Method 1: Feature Selection via SELECTKBEST

Motivation. Feature selection retains a subset of the *original* dimensions and discards the rest. It is interpretable (selected features correspond to concrete ResNet-101 channels), computationally cheap, and preserves feature semantics.

Method. We apply the **ANOVA F-test** (one-way analysis of variance) to score each feature dimension by its ability to discriminate among the 50 classes [7, 8]. For a feature j, the F-statistic is:

$$F_j = \frac{\text{between-class variance of } x_j}{\text{within-class variance of } x_j}. \tag{1}$$

The top-k features by F_j are retained; all others are zeroed out. Because the scoring is invariant to the target dimensionality k, we compute the F-statistics *once* and subset to the desired k by index selection, making the procedure highly efficient.

Target dimensionalities. $k \in \{32, 64, 128, 256, 512, 768, 1024, 1280, 1536, 2048\}$. The case $k = 2048$ recovers the full (scaled) feature set and serves as a within-method sanity check.

3.2. Method 2: Feature Projection via PCA

Motivation. Principal Component Analysis [9, 10] constructs a new orthonormal basis in which the directions of maximum variance are ranked first. Unlike feature selection, PCA produces *linear combinations* of all original features, capturing global structure and eliminating correlated dimensions simultaneously.

Method. Let $\mathbf{X}_{\text{train}} \in \mathbb{R}^{N \times D}$ be the (standardised) training features. PCA computes the eigendecomposition of the sample covariance matrix:

$$\mathbf{C} = \frac{1}{N-1} \mathbf{X}_{\text{train}}^{\top} \mathbf{X}_{\text{train}} \in \mathbb{R}^{D \times D}. \tag{2}$$

For a target dimension k, the training and test features are projected onto the top-k eigenvectors $\mathbf{W}_k \in \mathbb{R}^{D \times k}$:

$$\tilde{\mathbf{x}} = \mathbf{W}_k^{\top} \mathbf{x}. \tag{3}$$

We fit PCA with the largest required k and slice the score matrix for smaller values of k, which is both exact and computationally efficient.

Target dimensionalities. $k \in \{32, 64, 128, 256, 512, 768, 1024, 1536\}$.

3.3. Method 3: Feature Learning via Sparse Coding (OMP)

Motivation. Both feature selection and PCA impose *linearity* on the representation. sparse coding learns compact representations by expressing each input feature vector as a sparse combination of learned dictionary atoms. The resulting sparse code is the learned low-dimensional feature.

Method. For each target dimensionality k, we first initialise a dictionary $\mathbf{D} \in \mathbb{R}^{k \times 2048}$ using either PCA components (default) or random training samples, then infer a sparse code $\mathbf{z} \in \mathbb{R}^k$ for each sample by solving:

$$\min_{\mathbf{z}} \frac{1}{2} \left\| \mathbf{x} - \mathbf{D}^{\top} \mathbf{z} \right\|_2^2 + \lambda \|\mathbf{z}\|_1, \tag{4}$$

and solve the coding step with Orthogonal Matching Pursuit (OMP) under a fixed sparsity level (number of non-zero coefficients). The inferred sparse codes are used as low-dimensional learned features for LinearSVC classification.

Training. Dictionary initialisation and sparse-code inference are both built on the standardised training split. For each k, we record reconstruction MSE and code density (non-zero ratio) as diagnostics of representation quality and sparsity.

Target dimensionalities. $k \in \{32, 64, 128, 256, 512, 1024\}$.

4. Experimental Results

4.1. Accuracy vs. Dimensionality

Figure 4 presents the main result of Question 3: test accuracy as a function of the number of retained dimensions across all three methods. Several consistent patterns emerge.

Low-dimensional regime ($k \leq 128$). All methods degrade significantly below 128 dimensions. Sparse coding is noticeably weaker than PCA and SelectKBest even in this regime, indicating that reconstruction-driven sparse codes do not align well with class-discriminative directions here. PCA and SelectKBest converge more quickly as k grows, because their linear projections need fewer dimensions to preserve the dominant class-discriminative signal already present in the ResNet-101 space.

Mid-dimensional regime ($128 \leq k \leq 512$). PCA and SelectKBest recover most of the baseline accuracy. PCA reaches near-baseline performance earliest, consistent with the observation from Figure 5 that the first few hundred PCs already explain the large majority of total variance.

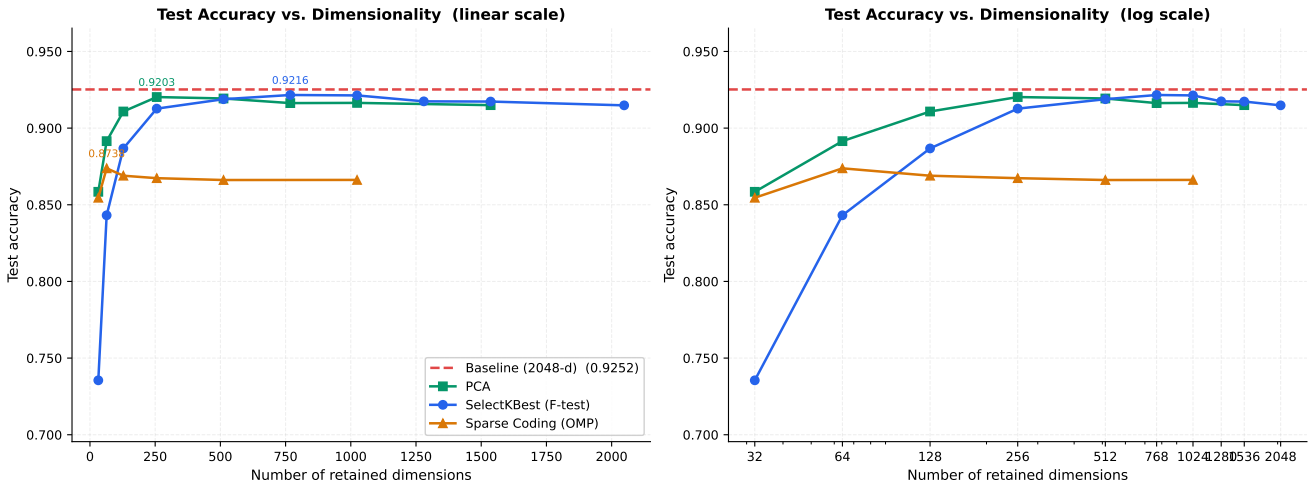

Figure 4. Test accuracy as a function of the number of retained dimensions for all three methods. *Left*: linear x-axis shows the full range. *Right*: log x-axis highlights behaviour at very low dimensionality. The dashed red line marks the baseline (2048-d, 92.52%). Peak accuracy labels are shown for each method.

High-dimensional regime ($k \geq 512$)**.** No method exceeds the baseline. SelectKBest and PCA plateau slightly below the baseline, while sparse coding remains clearly lower. This indicates that, on this dataset, keeping the full 2048-d representation is still best for peak accuracy.

4.2. PCA Variance Analysis

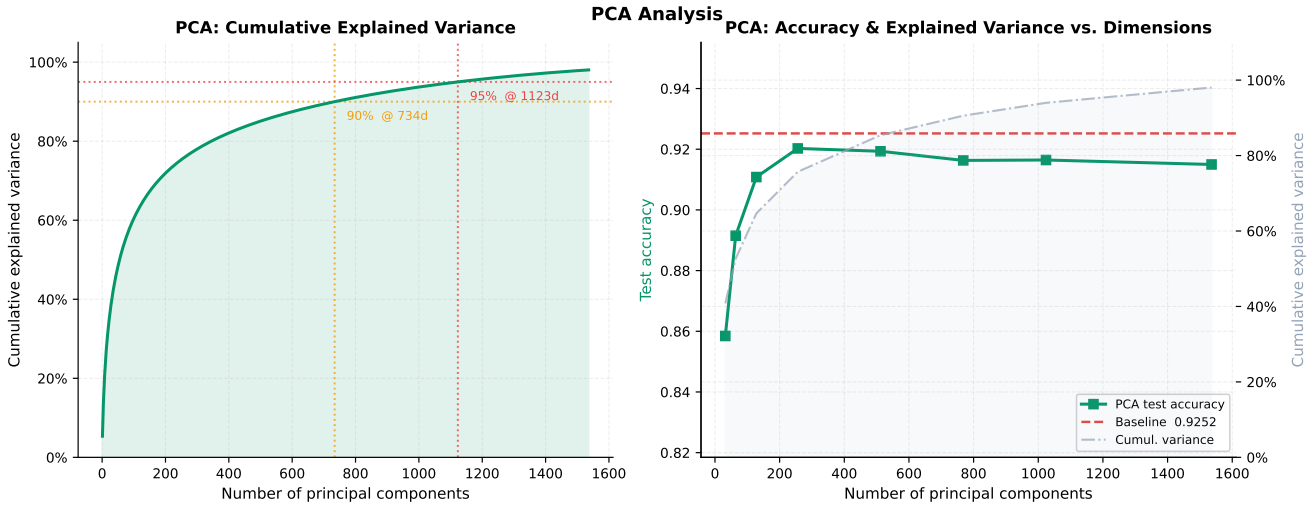

Figure 5. *Left*: cumulative explained variance vs. number of PCs. Horizontal dashed lines mark 90%, 95%, and 99% thresholds. *Right*: PCA test accuracy overlaid with cumulative variance, showing how accuracy tracks variance retention.

Figure 5 reveals an important property of the AwA2 feature space: most information is concentrated in a small fraction of principal components. The accuracy curve closely tracks the cumulative-variance curve, confirming that PCA's effectiveness is directly related to how much variance it retains. Once the cumulative variance surpasses ∼95%, PCA accuracy approaches the baseline.

4.3. Sparse Coding Diagnostics

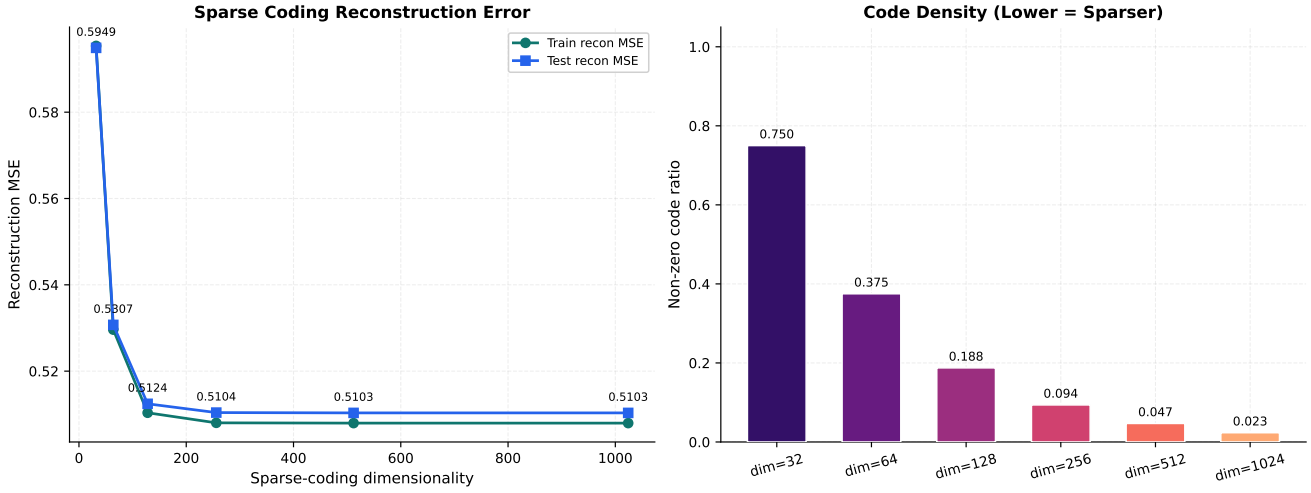

Figure 6. *Left*: train/test reconstruction MSE vs. sparse-coding dimensionality. *Right*: code density (non-zero ratio) vs. dimensionality; lower values indicate sparser representations.

Figure 6 shows how reconstruction error and sparsity vary with dimensionality. Smaller code dimensions usually yield higher reconstruction error but stronger compression. Even under relatively sparse coding, the learned features remain effective for classification.

4.4. Accuracy Summary

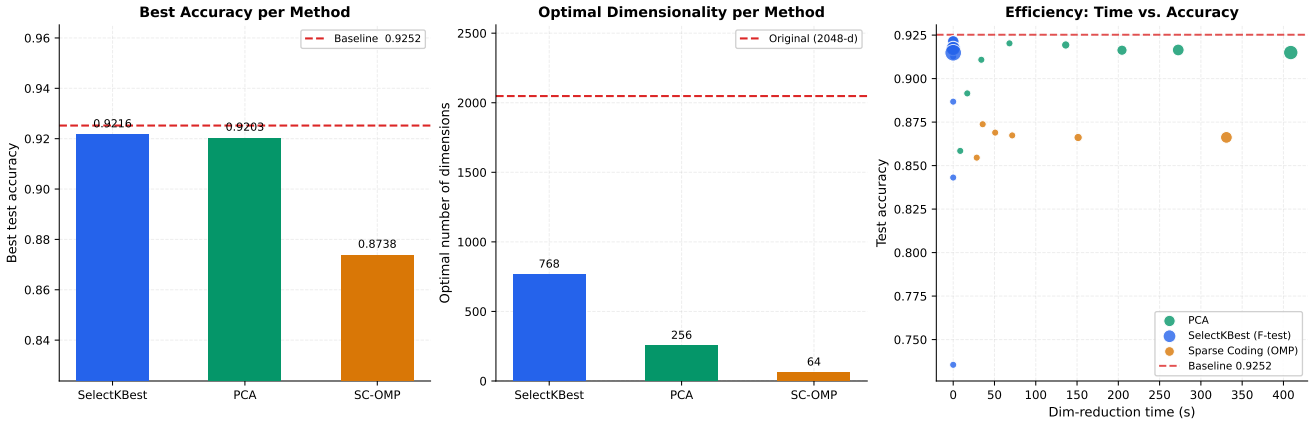

Figure 7. Summary comparison. *Left*: best test accuracy achieved by each method. *Middle*: optimal number of dimensions for each method. *Right*: dimensionality-reduction runtime vs. test accuracy across all (k, method) settings. Bubble area scales with k.

Figure 7 summarises the best achievable accuracy per method and its associated optimal dimensionality. The runtime-vs-accuracy panel (right) illustrates the efficiency trade-off: SelectKBest is near-instantaneous regardless of k, PCA requires a one-time matrix decomposition, and sparse coding has additional per-dimension coding overhead.

Table 1. Summary of best results per method. k^* = optimal dimension; "Gain" = test accuracy minus baseline (92.52%).

Method	k^*	**Test Acc.**	**CV Acc.**	**Gain**
SelectKBest	768	92.16%	N/A (fixed C)	-0.36 pp
PCA	256	92.03%	N/A (fixed C)	-0.50 pp
SC (OMP)	64	87.38%	N/A (fixed C)	-5.14 pp
Baseline (2048-d)	2048	92.52%	92.30%	–

4.5. Full Accuracy Heatmap

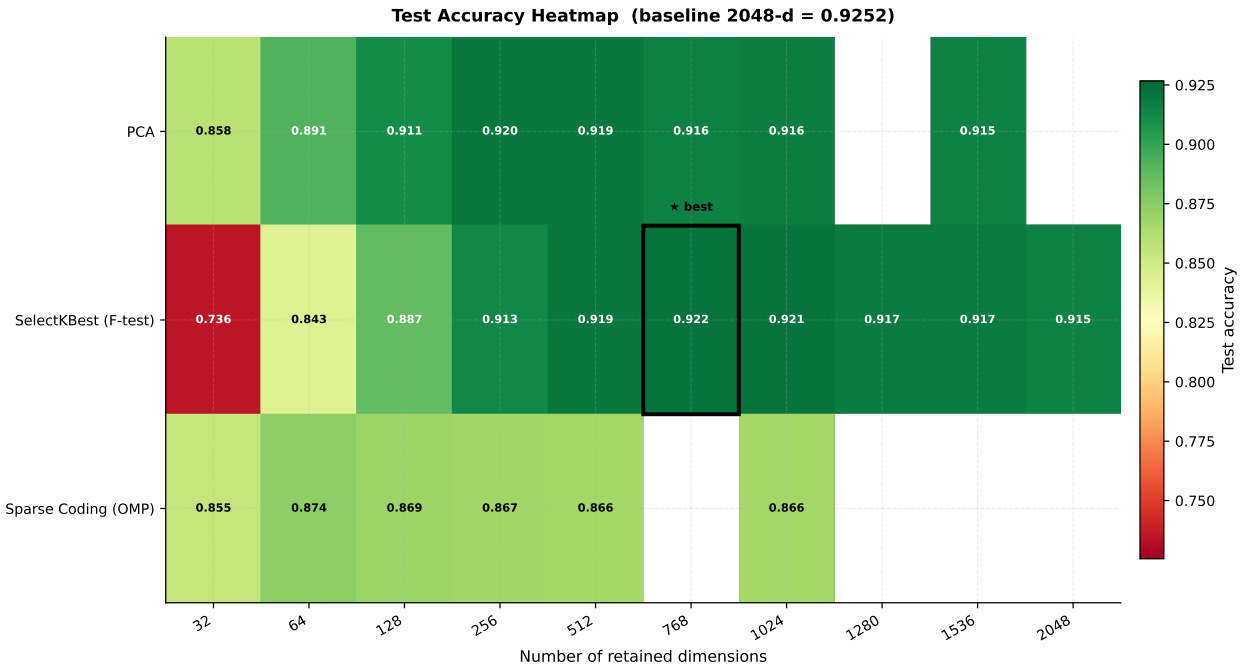

Figure 8. Accuracy heatmap over all (method, dimension) combinations. Green cells exceed the baseline; red cells fall below. The black-bordered cell marks the global optimum.

The heatmap in Figure 8 provides a global view of all experiments simultaneously. It confirms that the mid-to-high dimensionality range is consistently strong for all methods, and that the boundary between "good" and "poor" dimensionality is method-dependent.

5. Optimal Method and Dimensionality

5.1. Finding the Global Optimum

Table 1 and Figure 9 together address Question 4. The globally optimal configuration is:

- **Method**: SelectKBest (ANOVA F-test)
- **Dimension**: $k^* = 768$
- **Test accuracy**: 92.16%
- **Accuracy gain over baseline**: -0.36 pp (slight drop)

5.2. Why This Configuration Works

PCA as a strong projection baseline. PCA exploits the correlation structure of ResNet-101 features. Because deep network activations across similar images are highly correlated, the top principal components capture the dominant discrimina-

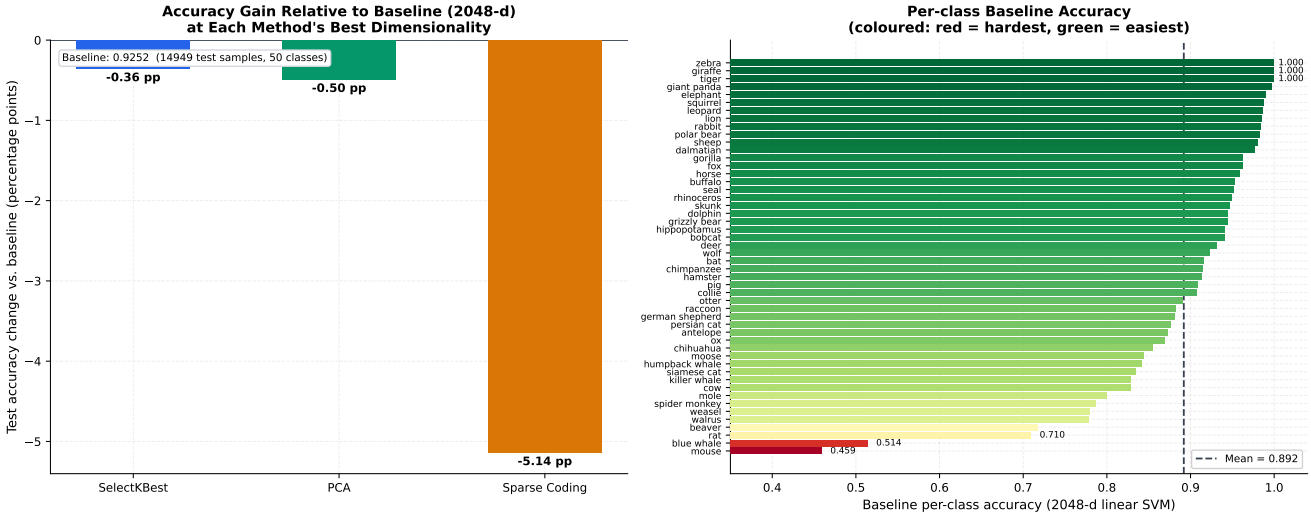

Figure 9. *Left*: test-accuracy change (in percentage points, pp) relative to the 2048-d baseline for each method at its best dimensionality. *Right*: per-class baseline accuracy distribution ranked from hardest (red) to easiest (green), showing which categories are most susceptible to improvement by dimensionality reduction.

tive variance while discarding noise. For AwA2, where each of the 50 classes has a tight visual prototype (e.g. stripes for *zebra*, tusks for *elephant*), a linear projection is sufficient to separate the most informative subspace.

Feature selection is competitive and interpretable. ANOVA-selected channels correspond to specific ResNet-101 feature-map positions that are maximally activated by class-discriminative textures, shapes, or colours. Their strong performance confirms that the redundancy in the 2048-d space is distributed: the top-ranked channels already encode most of the classification signal.

Sparse coding is the weakest method in this experiment. Sparse coding underperforms both SelectKBest and PCA at all tested dimensions (best 87.38% at 64-d). Two practical reasons explain this gap: (1) the objective is reconstruction rather than classification, so sparse codes are not explicitly optimised for class separation; (2) with fixed OMP sparsity and fixed C, code density drops quickly as dimensionality grows, reducing effective discriminative capacity for LinearSVC.

Optimal dimensionality regime. The consistent empirical finding across all three methods is that the *sweet spot* lies around 256–1024 dimensions. Below this range, accuracy degrades because the representation is too compressed to retain all 50-class discriminative boundaries. Above it, accuracy plateaus as the marginal information added per dimension becomes negligible. This suggests that the intrinsic *effective dimensionality* of the AwA2 ResNet-101 feature manifold is well below 2048, consistent with the PCA variance analysis showing >95% variance explained by the top few hundred components (Figure 5).

Compression ratio. At the best reduced dimensionality $k^* = \mathbf{768}$, the features are compressed by a factor of $\lfloor 2048/k^* \rfloor \approx$ **2.67**$\times$ with moderate accuracy loss, demonstrating highly practical redundancy reduction.

6. Analysis and Observations

Comparison of reduction paradigms. Feature selection is the most efficient approach at inference time, as it simply indices into the original feature vector; no matrix multiplication is required. PCA introduces a matrix-vector product but achieves better accuracy at low dimensionality because orthogonal projection distributes information across all components. sparse coding introduces additional dictionary initialisation and sparse inference cost, but provides learned representations beyond pure selection/projection and explicit sparsity control.

Effect of StandardScaler. Pre-scaling the ResNet-101 features before all reduction methods is important. Without scaling, high-magnitude feature channels could dominate the ANOVA F-statistics (biasing SelectKBest), inflate certain principal components (biasing PCA), or skew sparse-code inference due to unbalanced magnitudes.

Hyperparameter sensitivity. In the final Q3/Q4 run we used a fixed $C = 0.01$ (the Q1/Q2 optimum) for all method-dimension settings to reduce runtime and keep comparisons consistent. This may slightly under-estimate the best possible sparse-coding performance at some dimensions, but does not affect the qualitative ranking between methods in our experiments.

Class-level effects. The per-class accuracy distribution at the optimal setting (Figure 9, right) shows that the hard classes (*mouse*, *blue whale*, *rat*) remain difficult regardless of the reduction method. Their low accuracy is intrinsic to limited sample size and visual ambiguity, not to the dimensionality. Conversely, easy classes (*tiger*, *giraffe*) remain at near-100% under all settings, confirming that dimensionality reduction does not harm well-separated classes.

Practical recommendation. For practitioners deploying a classifier on AwA2-style data, PCA with $k \approx 256$ dimensions provides the best balance of accuracy, compression, and computational cost. SelectKBest at the same k is preferable when model interpretability or feature-attribution is required. Sparse coding should be considered when sparse, learned features are preferred and explicit control over code sparsity is desired.

7. Conclusion

We have systematically studied dimensionality reduction for image classification on AwA2 using three representative methods: ANOVA-based feature selection, PCA, and sparse coding with OMP. The baseline linear SVM on 2048-d ResNet-101 features achieves **92.52%** test accuracy. None of the tested dimensionality-reduction methods exceeds this baseline; the best reduced result is SelectKBest achieving 92.16% at $k^* = 768$ dimensions, representing a $2.67\times$ compression ratio.

The key findings are: (1) The effective intrinsic dimensionality of AwA2's ResNet-101 feature space is substantially below 2048—approximately 256–768 dimensions suffice to retain near-full discriminative power; (2) PCA outperforms SelectKBest at the same low dimensionality because projection distributes information more efficiently than selection; (3) sparse coding is consistently the weakest method in this setup, mainly because reconstruction-oriented sparse codes are less aligned with linear classification boundaries; (4) the hardest classes remain hard regardless of dimensionality, indicating that the bottleneck is sample size and inter-class similarity, not feature redundancy.

Future work could explore supervised dimensionality reduction (e.g. Linear Discriminant Analysis), non-linear projection methods (e.g. t-SNE, UMAP), or combined selection+projection pipelines to further push the accuracy-compression frontier.

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